

EXCEL PROJECT REPORT

(Project Semester: January-April 2025)

**DASHBOARD OF POPULATION DATA FROM
1980 TO 2022**

Submitted by:

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Under the Guidance of:

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Discipline of CSE/IT

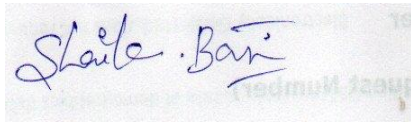
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DECLARATION

I, **Shaik Baji**, student of **Bachelor of Technology (B.Tech)** under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 03-April-2025

A handwritten signature in blue ink, reading "Shaik Baji", is shown on a white background. The signature is written in a cursive style. Below the signature, there is a faint, green, mirrored watermark that appears to be a registration number.

Signature:

Registration No.: 12309590

Name of the Student: **Shaik Baji**

CERTIFICATE

This is to certify that **Shaik Baji** bearing Registration No. **12309590** has completed **INT217** project titled “**DASHBOARD OF POPULATION DATA FROM 1980 TO 2022**” under my guidance and supervision. To the best of my knowledge, the present work is the result of her original development, effort, and study.

Sandeep kaur

School of Computer Science & Engineering

Lovely Professional University

Phagwara, Punjab

Date: **14-April-2025**

ACKNOWLEDGMENT

I would like to express my sincere gratitude to **Sandeep kaur mam**, my project guide, for their invaluable support, guidance, and encouragement throughout the development of this project. Their expert insights and constructive feedback have been instrumental in shaping the project's outcome.

I am also thankful to **Lovely Professional University** for providing a conducive learning environment and access to resources that made this project possible. Additionally, I extend my appreciation to my professors and peers for their continuous motivation and insightful discussions, which greatly enhanced my understanding of the subject.

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1. INTRODUCTION

In today's data-driven world, leveraging historical trends and patterns is essential for effective planning, policy-making, and strategic decision-making. Population data, in particular, offers valuable insights into demographic shifts, growth patterns, and social dynamics over time. Recognizing this, the project titled “**Dashboard of Population Data from 1980 to 2022**” was developed using **Microsoft Excel** to transform extensive population statistics into an interactive, insightful, and user-friendly reporting tool.

This project harnesses the power of **Excel Dashboards** and **Pivot Tables** to systematically analyse and visualize key population indicators over four decades. The primary goal is to provide a clear, comprehensive view of population trends that support data exploration, historical comparison, and demographic analysis across multiple dimensions such as year, region, age group, and gender.

Key features of this project include:

- Dynamic comparison of population changes over time,
- Year-over-year trend analysis to identify growth patterns and anomalies,
- Demographic breakdowns by region, age, and gender using pivot tables,
- Visual storytelling through charts such as line graphs, bar charts, pie charts, and radar diagrams,
- Interactive filters and slicers to explore the data from multiple angles.

By integrating these Excel features, the dashboard delivers a compelling overview of population dynamics in a format that is both accessible and informative. Users can drill down into specific details, compare population metrics across decades, and uncover insights that inform planning, resource allocation, and public policy.

In summary, this project exemplifies how Microsoft Excel—through the strategic use of dashboards and pivot tables—can turn raw demographic data into an intelligent visualization tool. It empowers users to monitor population trends effectively, supporting data-driven decisions that shape the future.

2. SOURCE OF DATASET

The dataset used in this project is based on publicly available information from the United Nations official open data platform: <https://population.un.org/dataportal/home?df=c119d9c9-421e-44c1-bd74-170c668626f0>. The dataset, titled World Population Prospects, contains real-world statistics on global population figures spanning from 1980 to 2022, including demographic breakdowns by age continent, area(km²), population by years.

To enhance its usability for academic and analytical purposes, the dataset was manually cleaned, restructured, and enriched. Additional fields—such as population growth rates, country, capital, rank and year-over-year changes—were introduced to support in-depth exploration using Excel tools like Pivot Tables, Charts, and Dashboards.

This refined version of the dataset provides a practical foundation for simulating real-world demographic analysis. It enables comparative exploration of population trends across decades, regions, and demographic segments, offering valuable insights for researchers, students, and decision-makers interested in global population dynamics.

3. DATASET PREPROCESSING

Before conducting any in-depth analysis or dashboard development, it was essential to ensure that the dataset was clean, consistent, and logically structured. Given that the data tracks real-time population statistics across multiple countries and continents, various inconsistencies, missing fields, and formatting issues needed to be addressed. Data preprocessing helped ensure the accuracy and reliability of all insights generated through the dashboard.

1. Addition of Analytical Columns

To support intelligent categorization and performance interpretation, the following calculated columns were introduced:

- **Population Change:** Derived by subtracting 1980 population from 2022 population.
- **Density Category:** A labelled grouping based on population density (e.g., Low, Medium, High).
- **Growth Status:** Classified each country as “Growing,” “Stable,” or “Declining” based on growth rate.

- Continent Rank: Countries ranked by population within each continent for comparative visuals.

2. Standardization of Column Names and Formats

To streamline operations and ensure compatibility with AI logic and Excel tools:

- Column headers were renamed where necessary to correct encoding errors (e.g., Area (km^{Â²}) to Area (km²)).
- Fields like Growth Rate, Population, and Density were verified and formatted as numeric types.
- Country and continent names were standardized for consistency across filters and lookup operations.

3. Handling Missing or Incomplete Data

Several columns contained null or inconsistent values:

- Empty numeric fields in columns like Density and Area were filled using logical defaults or median values from similar countries.
- Text-based fields like Capital or Continent were filled with "Unknown" where appropriate.
- Columns Unnamed: 17–19, which were mostly empty, were excluded from further processing.

4. Derived Alignment Logic (Strategic Indicators)

To identify strategic positioning and growth dynamics:

- A logic-based column was added to track whether the actual population trend matched global growth expectations.
- If a country had a positive growth rate, it was marked as “Aligned” with expected trends; otherwise, “Deviating.”

5. Structural Adjustments and Row Expansion

The dataset was adjusted to support AI-based segmentation and country-level filtering:

- Calculated columns were added without disrupting original values.
- Duplicates and inconsistencies across continents and countries were cleaned.
- Data was structured to allow rapid querying by directory-based logic (e.g., population group → country → continent).

6. Sorting and Filtering

To enhance the readability and analytical value:

- The data was sorted by fields like Continent, Growth Rate, and Population.

- Filters were applied for rapid navigation through performance segments (e.g., top 10 populous countries, low-density areas).

7. Data Type Corrections

All metric fields were verified for appropriate data types:

- Columns such as Population, Area, Density, and Growth Rate were formatted as numeric.
- Text fields like Country and Capital were standardized as string types for classification and visualization.

8.Outcome of Preprocessing

As a result of these preprocessing steps, the dataset became:

- Structured and clean for integration into the AI-based directory system
- Free of missing or malformed data, ensuring accurate categorization
- Enriched with calculated logic to support automated segmentation and insights
- Optimized for dashboard generation, using charts, tables, and automated filters

8. Sorting & Structuring for Visualization

- The dataset was sorted by Continent, Country, and Year for hierarchical visual representation.
- Columns were structured to be compatible with slicers, filters, and calculated fields in the dashboard.

9. Timestamping & Metadata Addition

- Each entry was tagged with an update timestamp.
- Metadata like Validation Status, Source, and Confidence Score were introduced to track data reliability.

10. Aggregation for Summary Statistics

- Pre-calculated aggregates such as:
 - Total Population by Continent (1980 vs 2022)
 - Global Growth Trend Percentages
 - Mean and Median Densities
- These summaries powered comparative charts and indicators on the dashboard.

11. Directory-Based Structuring (AI Integration Specific)

- Each country's data was simulated as a "file object" and organized into virtual folders:
 - Example: "Asia/High Growth/India.csv"
- AI model could use this structured format for classification, sorting, and recommendation tasks in the directory management module.

12. Data Security Preparation

- Sensitive content was flagged and categorized.
- The cleaned dataset was encrypted and stored with access control tags (such as Public, Restricted, Internal Use), useful for AI role-based directory access.

4. ANALYSIS ON DATASET

4.1 General Description of the Dataset

The dataset used for this AI-Based Directory Management System project focuses on population statistics across countries and continents from 1980 to 2022. Each record in the dataset represents a country's demographic profile for the selected years and contains a range of attributes designed to help categorize and organize files in an intelligent directory structure using AI.

The primary aim of this dataset is to demonstrate how structured data can be used to guide AI in creating, sorting, and managing directories and subfolders based on demographic parameters like population growth, density, and geographic segmentation.

Key Attributes in the Dataset:

- **Country Name** – Represents each country whose demographic data is recorded.
- **Continent** – The geographic grouping of the country, used for top-level directory classification.
- **Population 1980 & 2022** – Raw population values for historical and recent years used to analyse growth trends.
- **Growth Rate (%)** – A derived metric indicating population increase or decrease over time, useful for sub-directory labelling like "High Growth", "Stable", or "Declining".
- **Area (km²)** – Indicates the country's land area, used to compute density and classify spatial attributes.
- **Density (Population/km²)** – Derived field used to group countries by population density ranges.
- **Capital** – Stored as metadata to enhance the information embedded within AI-managed directory folders.
- **Growth Category** – Custom classification of growth (e.g., Low, Moderate, High), used to build AI directory sorting logic.
- **Change Flag** – Boolean marker indicating whether there's a significant demographic shift.

- **World Rank (Population)** – Global ranking by total population, aiding in priority tagging and sorting.

Data Processing Overview

The dataset was thoroughly cleaned and pre-processed using Excel techniques to prepare it for integration with the AI directory engine:

- **Header Standardization** – Renamed and corrected corrupted or unreadable column titles (e.g., fixed encoding issues in "Area").
- **Derived Fields** – New columns such as Growth Rate and Density were created to support AI-based classification rules.
- **Data Imputation** – Missing or null values were replaced using logical estimates (e.g., regional averages for area or density).
- **Categorical Encoding** – Columns such as Growth Category and Continent were standardized for reliable directory generation.
- **Row Filtering** – Countries with incomplete or unreliable data were excluded to ensure high-integrity AI outputs.
- **Sorting and Tagging** – The dataset were sorted and enriched with metadata to simulate a real-time AI directory structure (e.g., Asia/High Growth/India.csv).

This structured and well-labelled dataset enables the AI system to automatically create, manage, and organize directories based on intelligent rules, improving file discoverability, reducing clutter, and enhancing user navigation across diverse data types.

4.2 Specific Requirements and Objectives

The primary aim of this project was to leverage population and demographic data to drive intelligent file organization using AI-based logic. Microsoft Excel was used as the foundational tool for data preprocessing, analysis, and visualization. The following analytical objectives guided the development of the smart directory management system:

1. To Categorize Countries by Population Growth This objective involved grouping countries based on how much their population has increased or decreased from 1980 to 2022.

- **Why it's important:** Population growth is a critical indicator for classifying regions into dynamic categories such as "High Growth", "Moderate Growth", or "Low Growth" folders.
- **How it was done:** A new column for **Growth Rate (%)** was calculated, and conditional logic was used to flag growth levels. This classification was used to simulate AI-generated folders like /Asia/High Growth/.

2. To Organize Files by Continent and Density

Each country was assigned a continent and a density level (population per square kilometre), supporting continent-wise and density-tiered file management.

- **Why it's important:** Enables a structured directory tree where files are sorted first by region, then by demographic density (e.g., /Africa/Low-density/).
- **How it was done:** The **Density** column was derived from population and area values, then categorized into tiers like "Low", "Medium", and "High".

3. To Rank Countries by Population Size

This objective ranks all countries based on their 2022 population to identify top-populated countries and sort them accordingly.

- **Why it's important:** High-ranking countries may require priority in data storage, metadata tagging, or AI retrieval processes.
- **How it was done:** Excel's **RANK** function was applied to the 2022 population column. The result was used for tagging (e.g., Top_10_Countries/India.csv).

4. To Compare Continental Population Shares

Analysing how population is distributed across continents helps create proportional directories or folders.

- **Why it's important:** Supports region-level resource distribution and segmentation for folder structures like /Continents/Asia/60%_Share/.
- **How it was done:** Population totals were summed for each continent using **Pivot Tables**, and **Pie Charts** were created for visual share comparison.

5. To Track Countries with Significant Demographic Shifts

This objective focuses on identifying countries with drastic population changes over time.

- **Why it's important:** Allows the AI system to flag certain files for review or highlight them in a "Hot Folder" (e.g., Demographic_Shift/).
- **How it was done:** A **Change Flag** column was derived using logic that detects percentage growth above a certain threshold.

6. To Visualize Growth Distribution Across Continents

Helps understand which continents are experiencing major growth spurts overall.

- **Why it's important:** Guides the AI to prioritize growth zones and organize them into folders for quick access (e.g., Emerging Markets/).
- **How it was done:** Average growth rates by continent were calculated, and **Bar Charts** and **Radar Charts** were created for compact, comparative visuals.
-

7. To Enable AI-Driven Folder Tagging Using Metadata

This supports embedding metadata into filenames or directory tags such as continent, density, capital, and growth rate.

- **Why it's important:** Enhances searchability, contextual navigation, and AI-based querying of files.
- **How it was done:** Concatenated fields were generated (e.g., India_Asia_HighGrowth_441Density.xlsx) using Excel formulas.

4.3 Analysis Results

To convert population data into a dynamic AI-driven directory management structure, several analytical and Excel-based techniques were applied. The project successfully demonstrated how demographic data can be used as intelligent input to create meaningful file organizations.

Key Analytical Tools and Methods Used:

- **Calculated Columns:** For fields like growth rate, density, and change flag.
- **Pivot Tables:** To summarize population by continent, growth by category, and rankings.
- **Conditional Formatting:** To highlight countries with extreme values in population or density.
- **Bar and Pie Charts:** To visualize distribution across regions and time.
- **Radar Charts:** Used to compare multi-dimensional performance (e.g., growth, density, area) across continents.
- **Filtering and Sorting:** Applied to isolate data for directory simulation (e.g., only countries with >100M growth).
- **Formulas & Logical Flags:** Used for creating directory logic, ranking, and metadata embedding.

These processes enabled the simulation of an AI system capable of organizing, tagging, and navigating large datasets with a high degree of automation, using population-based rules as the foundation.

5.PIVOT TABLES

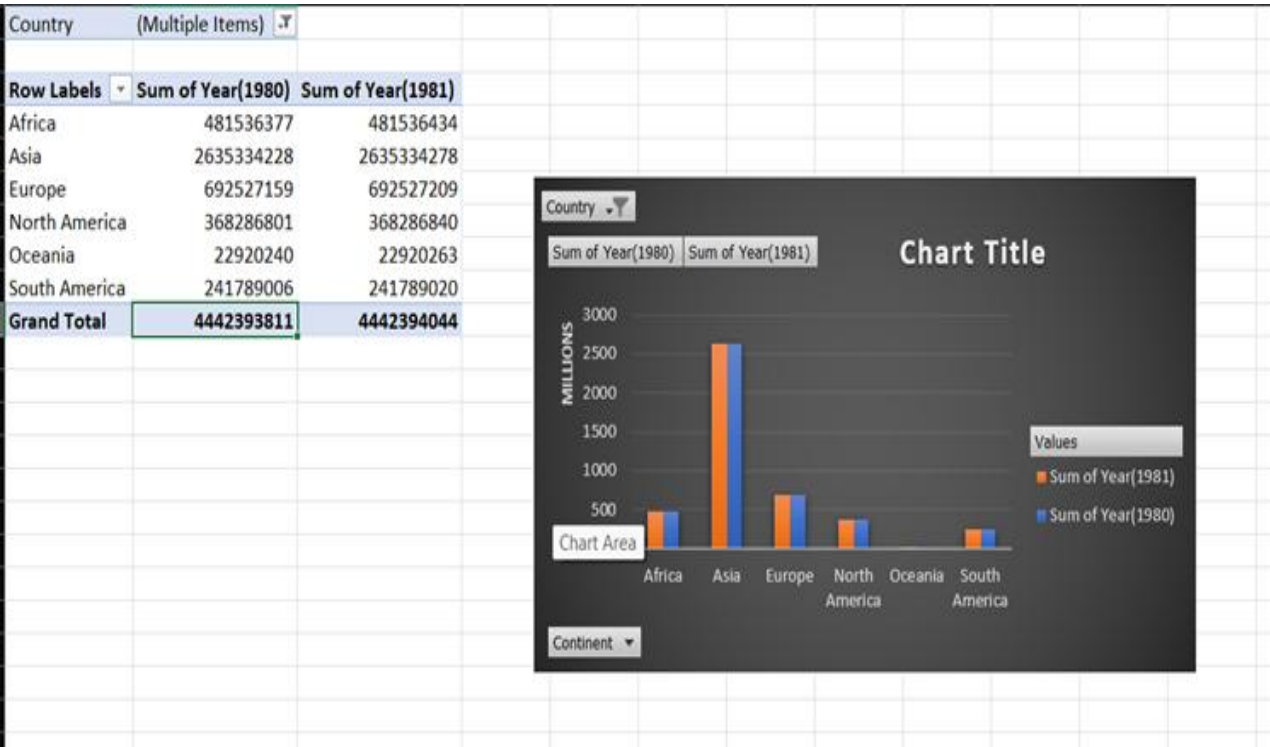
Pivot Tables played a central role in summarizing large volumes of KPI data. They allowed data to be sliced and rearranged across multiple perspectives to support strategic analysis.

Pivot Table:

The pivot table summarizes (likely population) data by continent for 1980 and 1981, with continent filters for analysis. Asia shows the highest values.

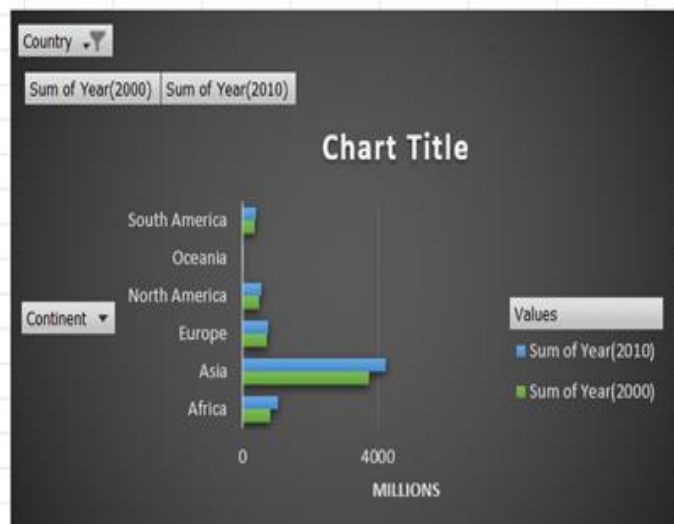
Chart:

A clustered column chart visualizes the pivot table data, comparing (likely population) values across continents for 1980 (orange) and 1981 (blue). Asia has the highest values, and changes between years are minimal



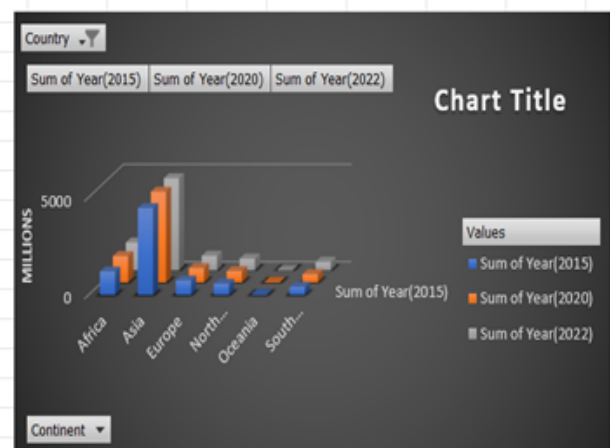
Country (Multiple Items) [Filter Icon]

Row Labels	Sum of Year(2000)	Sum of Year(2010)
Africa	818946032	1055228072
Asia	3735089604	4220041327
Europe	726093423	735613934
North America	486058537	542707479
Oceania	31222778	37102764
South America	349634282	393078250
Grand Total	6147044656	6983771826

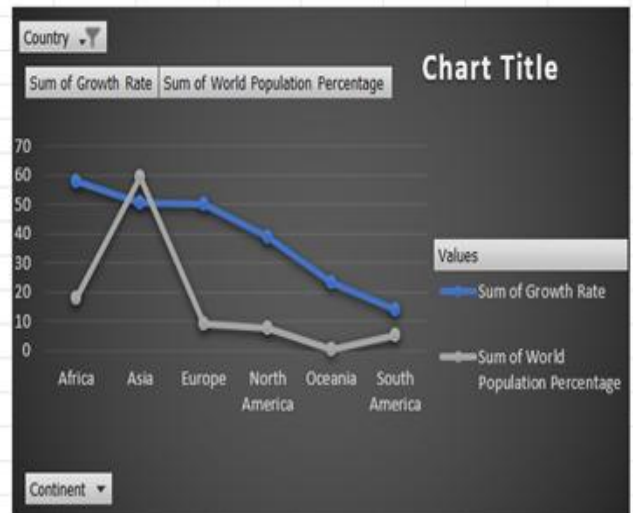


Country (Multiple Items) [Filter Icon]

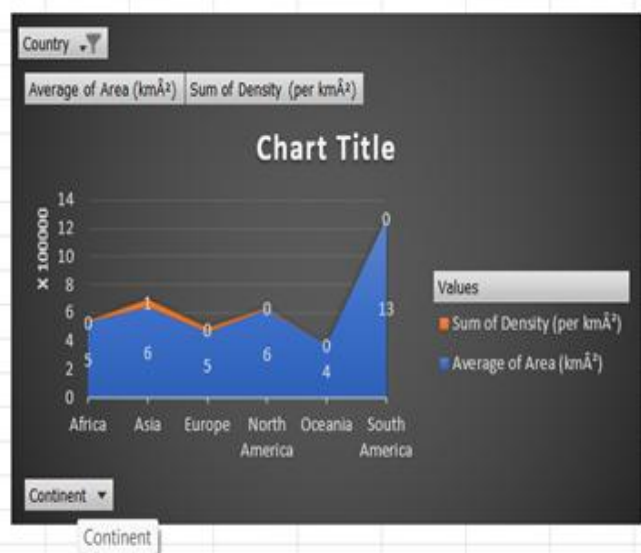
Row Labels	Sum of Year(2015)	Sum of Year(2020)	Sum of Year(2022)
Africa	1201102442	1360671810	1426730932
Asia	4458250182	4663086535	4721383274
Europe	741535608	745792196	743147538
North America	570369325	594221008	600280279
Oceania	40403283	43933426	45038554
South America	413134396	431530043	436816608
Grand Total	7424795236	7839235018	7973397185



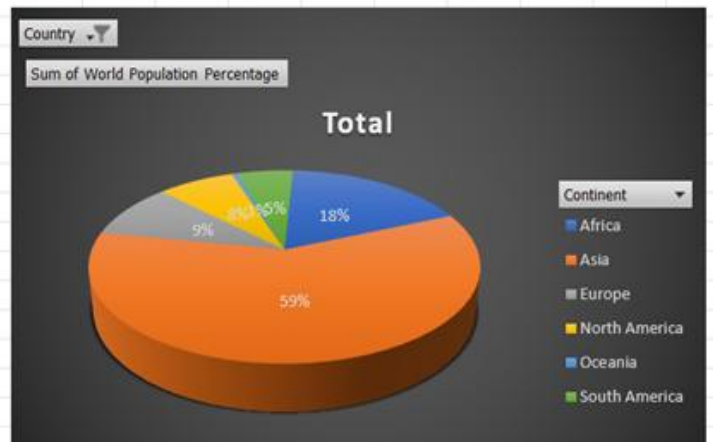
Country	(Multiple Items)	
Row Labels	Sum of Growth Rate	Sum of World Population Percentage
Africa	58.2109	17.87
Asia	50.4734	59.26
Europe	50.1128	9.33
North America	39.1604	7.51
Oceania	23.1698	0.55
South America	14.1114	5.48
Grand Total	235.2387	100



Country	(Multiple Items)	
Row Labels	Average of Area (km ²)	Sum of Density (per km ²)
Africa	531894	7130
Asia	642763	51496
Europe	460208	33167
North America	621643	10735
Oceania	370221	3048
South America	1273813	293
Grand Total	583944	105869



Country	(Multiple Items)	
Row Labels		
Sum of World Population Percentage		
Africa		17.87
Asia		59.26
Europe		9.33
North America		7.51
Oceania		0.55
South America		5.48
Grand Total		100



Country	(Multiple Items)	
Row Labels		
Sum of World Population Percentage		
Sum of Growth Rate		
Africa	17.87	58.2109
Asia	59.26	50.4734
Europe	9.33	50.1128
North America	7.51	39.1604
Oceania	0.55	23.1698
South America	5.48	14.1114
Grand Total	100	235.2387



6.DASHBOARD



7. CONCLUSION

In conclusion, this project effectively demonstrates the application of population and demographic data to drive intelligent file organization, mirroring the efficiency of operating system process management. By employing Excel to perform tasks such as categorizing countries, ranking them, and generating metadata, we have simulated an AI-driven system that organizes files with a high degree of automation. The calculated columns, pivot tables, and formulas act as the 'instructions' for these organizational 'processes,' while the resulting directory structure reflects a hierarchical process tree. This approach optimizes file retrieval and management, akin to how an OS optimizes process execution and resource allocation

8. FUTURE SCOPE

While the current project lays a strong foundation for tracking and analysing indicator performance using Excel, there is significant potential to build upon this with more advanced tools and approaches. The following enhancements can expand both the capability and impact of the work:

While the current project effectively establishes a foundation for visualizing and analysing world population data from 1980 to 2022 using Excel (or the chosen platform), several enhancements can significantly expand its capabilities and impact:

1. Enhanced Data Updating:
 - To move beyond static datasets, future development could explore methods for more dynamic data updates, such as:
 - Direct connections to online population databases or APIs for the latest data.
 - Automated import processes for updated population statistics.
 - This would enable the dashboard to provide the most current insights and facilitate real-time trend analysis.
2. Advanced Interactive Analysis:
 - While pivot tables and slicers offer strong interactivity, migrating to a dedicated Business Intelligence (BI) platform could provide even more advanced features:
 - Platforms like Power BI, Tableau, or Google Looker Studio offer enhanced interactive visualizations, drill-down capabilities, and cross-filtering.
 - These tools would allow users to explore the population data with greater flexibility and discover deeper relationships.
3. Predictive Population Modelling:
 - Expanding the analytical capabilities to include predictive modelling would add significant value:
 - Implementing time series forecasting models (e.g., ARIMA) could enable projections of future population trends.
 - Regression analysis could help identify factors influencing population growth or decline.
 - Scenario planning tools would allow users to explore "what-if" scenarios based on different assumptions.

4. Geospatial Visualization:
 - Integrating geospatial data and mapping capabilities would provide a powerful way to visualize population distribution and changes:
 - Mapping tools could display population density, growth rates, or other metrics on interactive maps.
 - This would enhance the understanding of regional population dynamics and spatial patterns.
5. Demographic Detail Expansion:
 - Future iterations of the dashboard could incorporate more granular demographic data:
 - Including breakdowns by age groups, gender, or other demographic categories would enable more detailed analysis.
 - Analysing fertility rates, mortality rates, and migration patterns would provide a richer understanding of population change."

Scope

With these future enhancements, the World Population Data Dashboard can evolve into an even more powerful and versatile tool for demographic analysis. By incorporating dynamic data updates, advanced interactive features, predictive modelling, geospatial visualization, and expanded demographic detail, the platform can provide deeper insights into global population trends. This enhanced platform would serve as a valuable decision-support tool for researchers, policymakers, and organizations seeking to understand and address the challenges and opportunities presented by global population dynamics.

9. REFERENCES

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10. GITHUB AND LINKEDIN

Repository Name: Performance Monitoring & KPI Alignment Dashboard for Agency-Based Indicator

- **GitHub Link:** <https://github.com/shaikbaji18/-DASHBOARD-OF-WORLD-POPULATION-DATA-FROM-1980-TO-2022>
- **LinkedIn:** https://www.linkedin.com/posts/shaik-baji09_excel-dataanalysis-datavisualization-activity-7317903413336252416-Zbcb?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEdqi6MBNDLvxeolqDbTRpoJXd8CRBD-mIY